Surgical skill assessment from kinematics and video

Novice / intermediate / expert on JIGSAWS Suturing under leave-one-user-out

Group 17

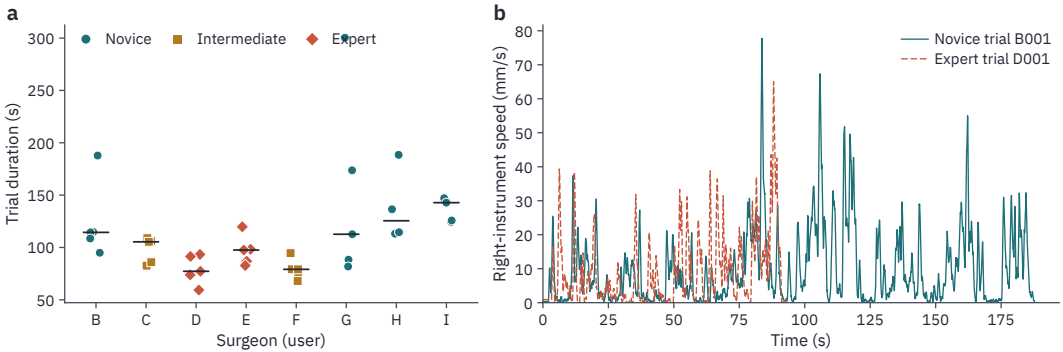
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Why automate skill assessment?

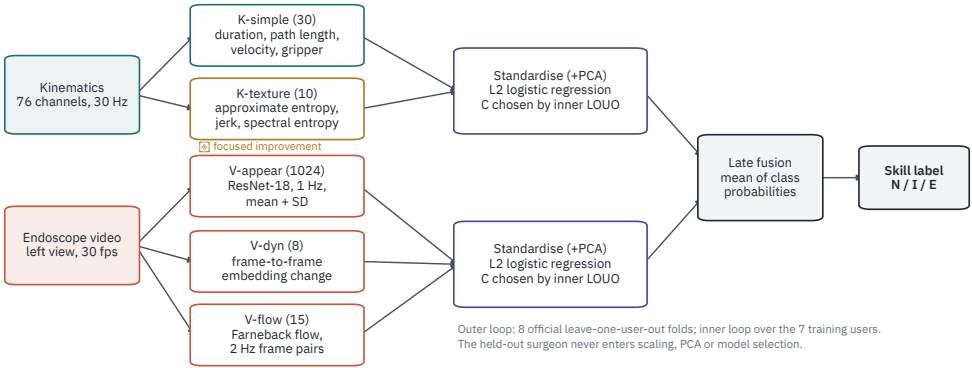
- Surgical skill predicts outcomes: the lowest-rated quartile of surgeons had far more complications than the highest-rated one (Birkmeyer et al., 2013).
- Expert rating with OSATS is slow, costly and partly subjective.
- A surgical robot already logs every instrument movement and the endoscope video, so assessment can use data that are collected anyway.
- A useful system must work for a surgeon it has never seen, so we hold out whole surgeons (LOUO), not single trials.

Task and data

JIGSAWS Suturing: 39 trials from 8 surgeons; label = self-reported level, N (4 surgeons, 19 trials), I (2, 10), E (2, 10). 76 kinematic channels at 30 Hz and endoscope video. Official LOUO folds hold out one surgeon, hence one class. Only two surgeons per trained class.



Pipeline



Every block $\rightarrow$ standardise ($\rightarrow$ PCA for V-appear) $\rightarrow$ L2 logistic regression; C chosen by an inner LOUO loop on the training surgeons only. Late fusion = mean of class probabilities.

Features

Kinematics (patient-side instruments)

- **K-simple (30)**: path length, speed, angular speed, idle fraction, economy, gripper statistics, duration, bimanual coordination.
- **K-texture (10)**, focused improvement: approximate entropy, log dimensionless jerk, spectral entropy. Idea from Zia & Essa: experts move more regularly.

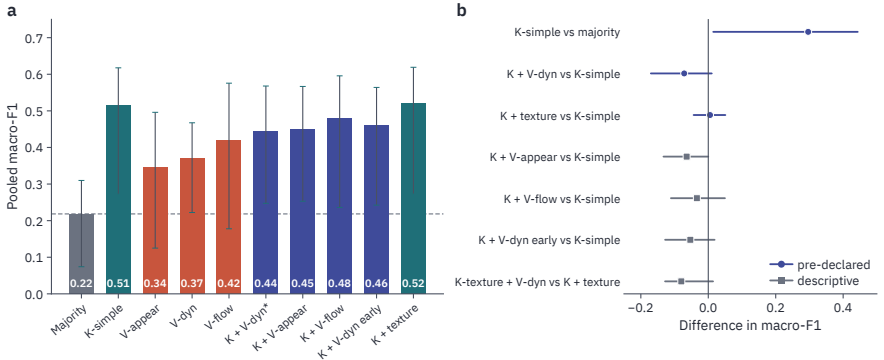
Video (left camera)

- **V-appear**: frozen ResNet-18 at 1 Hz, mean + SD of embeddings, in-fold PCA.
- **V-dyn (8)**: how fast consecutive embeddings change; cancels static appearance of a session.
- **V-flow (15)**: Farneback optical flow at 2 Hz.

Evaluation protocol

- Metric: **pooled macro-F1** over the 39 out-of-fold predictions (per-fold macro-F1 is undefined with one class per fold).
- Uncertainty: **user-level bootstrap** (2000 resamples of surgeons) and an **exact permutation test** over all $8!/(4!\,2!\,2!) = 420$ label assignments.
- Pre-registered in `PROTOCOL.md`: all models and three comparisons fixed before the run; every model evaluated once.
- **Shortcut audit**: surgeon-identity probe per block; duration-only classifier; features residualised on log duration.

Results



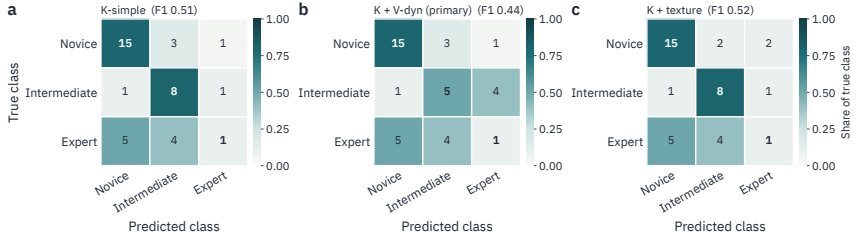
Majority 0.218 | **K-simple 0.515** ($p = 0.033$) | K+V-dyn (primary) 0.443 | K-simple+texture 0.520

The three pre-declared comparisons

Comparison	Δ macro-F1	95% user bootstrap	Verdict
K-simple vs majority	$+0.296$	$[+0.015, +0.443]$	**supported**
+ video (V-dyn) vs K-simple	-0.071	$[-0.170, +0.010]$	not distinguishable
+ texture vs K-simple	$+0.005$	$[-0.043, +0.051]$	not distinguishable

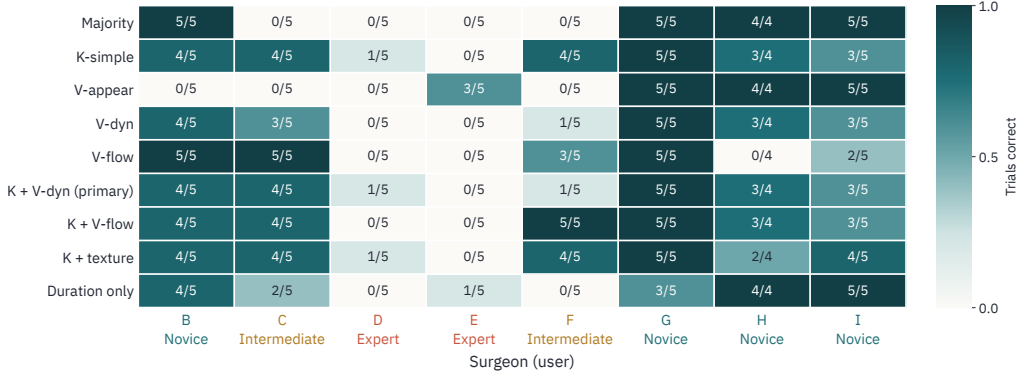
- Video alone is not above chance (0.345, 0.371, 0.420).
- Every fusion variant scored below kinematics alone.

Where it fails: the experts



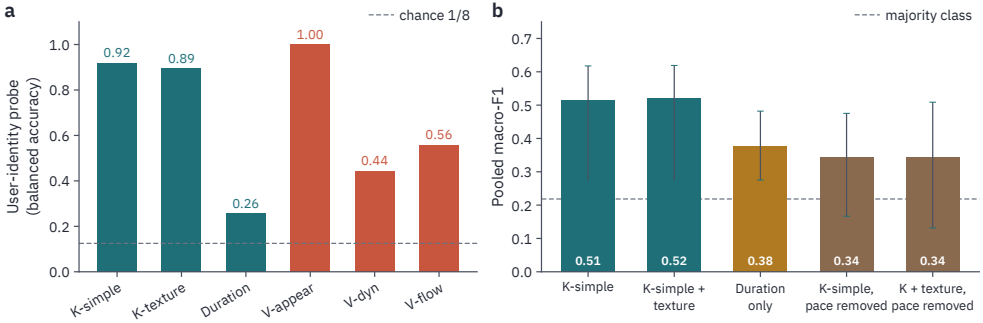
- Novices F1 0.75, intermediates 0.64, experts only 0.15: 5 of 10 expert trials called N, 4 called I.
- Holding out one expert leaves one expert in training, so the model learns that person rather than expertise.
- Experts work at the pace of intermediates (median 89 s vs 84 s; novices 124 s).

Per-surgeon results



Experts D and E fail for every kinematic model. Fusion hurts intermediate F (K-simple 4/5 → fused 1/5). V-appear is right or wrong for whole surgeons, as an identity detector would be.

Shortcut audit: identity and pace



- Surgeon identity from one trial: K-simple 0.92, V-appear 1.00 (chance 0.125).
- Duration alone reaches 0.375; removing log duration from K-simple drops it from 0.515 to 0.342.
- Texture failed: log jerk vs log duration $|r| = 0.93$, identity 0.89. V-dyn mostly repeats pace ($|r|$ 0.54).

How this differs from prior work

- Strict unseen-surgeon evaluation with macro-F1 and explicit uncertainty; many published near-100% results use LOSO.
- Zia & Essa's 88.2% LOUO accuracy cannot be reproduced as a pooled count over 39 trials, so it is not directly comparable.
- We **measure** identity and pace shortcuts instead of assuming a high score means skill.
- Pre-registration and an exact permutation test make negative results credible.

Conclusion

- Simple kinematic statistics transfer to new surgeons (0.515 vs 0.218), mostly through pace.
- Whole-frame video and simple fusion did not help; motion texture did not help.
- Experts cannot be learned from one training expert.
- Next: more surgeons (all JIGSAWS tasks, other datasets), pace-independent features, instrument-region video, fusion weights learned in-fold.

AI tools: Claude Code (Claude Fable 5.1, Claude Opus 5.5) helped plan, code, run and write; the group reviewed everything and is responsible for the work.



Thank you

Questions?